

Section 4.5 Curve Sketching

59) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values. Also, find any asymptotes the function may have.

$$y = \frac{x^2}{x^2 + 4}$$

$$y' = \frac{2x(x^2 + 4) - x^2(2x)}{(x^2 + 4)^2}$$

$$x = \pm 1.155$$

Concave up $(-1.155, 1.155)$

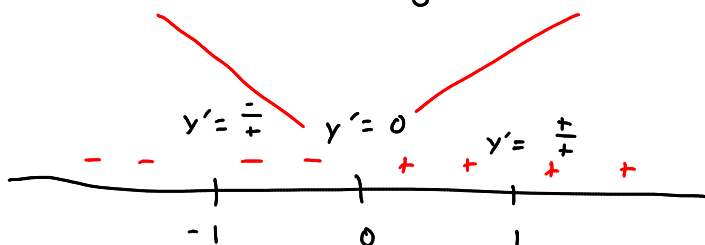
Concave Down $(-\infty, -1.155) \cup (1.155, \infty)$

$$y' = \frac{8x}{(x^2 + 4)^2} = 0$$

$$\begin{aligned} y' &= 0 \\ 8x &= 0 \\ x &= 0 \end{aligned}$$

$$\begin{aligned} y' &\text{ DNE} \\ (x^2 + 4)^2 &= 0 \\ x^2 + 4 &= 0 \\ \sqrt{x^2} &= \sqrt{-4} \end{aligned}$$

imaginary number



Increasing $(0, \infty)$

Decreasing $(-\infty, 0)$

Min $(0, 0)$

NO Maximum

$$y'' = \frac{8(x^2 + 4)^{-2} - 8x \cdot 2(x^2 + 4)^{-3} \cdot 2x}{(x^2 + 4)^4}$$

$$= \frac{8x^2 + 32 - 32x^2}{(x^2 + 4)^3}$$

$$= \frac{-24x^2 + 32}{(x^2 + 4)^3} = 0$$

$$-24x^2 + 32 = 0$$

$$-24x^2 = -32$$

$$x^2 = \frac{32}{24}$$

$$x = \pm \sqrt{\frac{32}{24}}$$

Horizontal Asymptote

$$\lim_{x \rightarrow \pm \infty} \frac{x^2}{x^2 + 4}$$

$$\lim_{x \rightarrow \pm \infty} \frac{2x}{2x} = 1$$

H. Asymptote at $y = 1$

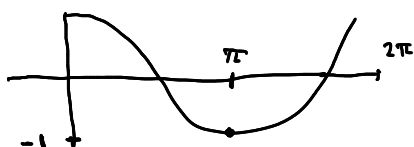
60) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values. Also, find any asymptotes the function may have.

$$y = x + \sin x \quad [0, 2\pi]$$

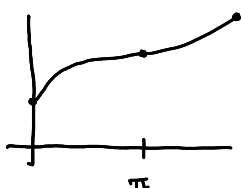
$$y' = 1 + \cos x$$

$$0 = 1 + \cos x$$

$$-1 = \cos x$$



$$x = \pi$$



incr $[0, \pi) \cup (\pi, 2\pi]$

Decr none

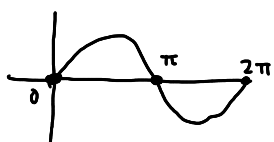
Min $(0, 0)$

Max $(2\pi, 2\pi)$

$$y'' = -\sin x$$

$$0 = -\sin x$$

$$0 = \sin x$$



Concave up $\cap$ $(\pi, 2\pi)$

Concave Down $\cup$ $(0, \pi)$

No Asymptote

61) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values. Also, find any asymptotes the function may have.

$$y = \frac{\ln x}{x}$$

$$y' = \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2}$$

$$y' = \frac{1 - \ln x}{x^2} = 0$$

$$y' = 0$$

$$1 - \ln x = 0$$

$$1 = \ln x$$

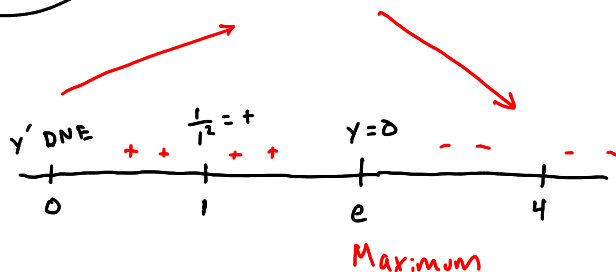
e

$$e = x$$

$$y' \text{ DNE}$$

$$x^2 = 0$$

$$x = 0$$



INCR $(0, e)$

DECR (e, ∞)

max $(e, 0.368)$

No minimums

Vasym $x = 0$

CD $(0, 4.482)$

CU $(4.482, \infty)$

$$y' = \frac{1 - \ln x}{x^2}$$

$$y'' = \frac{-\frac{1}{2} \cdot x^2 - (1 - \ln x) \cdot 2x}{x^4}$$

$$y'' = \frac{-x - 2x(1 - \ln x)}{x^4}$$

$$y'' = \frac{-x [1 + 2(1 - \ln x)]}{x^4} = 0$$

$$-x = 0$$

or

$$1 + 2(1 - \ln x) = 0$$

$$1 + 2 - 2\ln x = 0$$

$$3 = 2\ln x$$

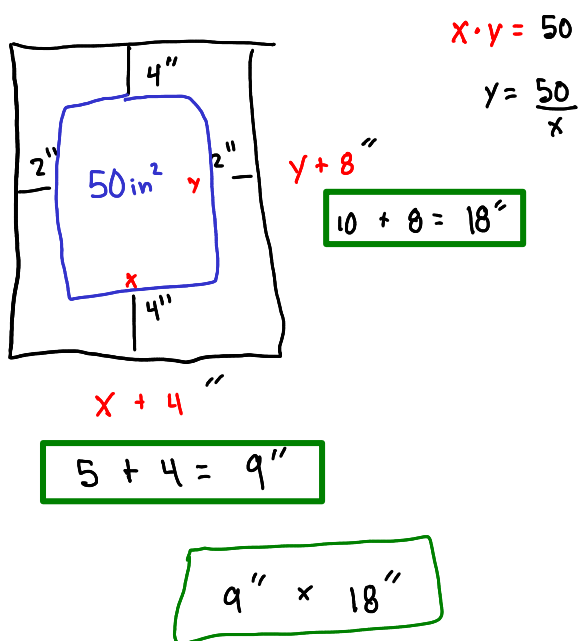
$$\frac{3}{2} = \ln x$$

$$x = e^{\frac{3}{2}} = 4.482$$



Section 4.7 Optimization Problems

62) *Designing a poster.* You are designing a rectangular poster to contain 50 in^2 of printing with a 4-inch margin at the top and bottom and a 2 inch margin at each side. What are the dimensions that will minimize the amount of paper used?



$$P = (x+4)(y+8)$$
$$P = (x+4)\left(\frac{50}{x} + 8\right)$$
$$P = 50 + 8x + \frac{200}{x} + 32$$

$$P' = 8 - \frac{200}{x^2} = 0$$

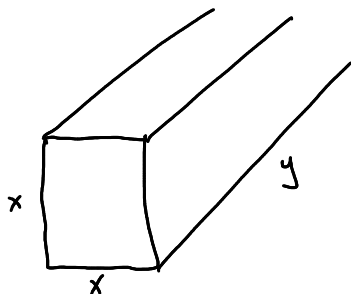
$$\frac{200}{x^2} = 8$$

$$+ \sqrt{\frac{200}{8}} = \sqrt{x^2}$$

$$5'' = x \quad \text{inside}$$

$$y = \frac{50}{5} = 10'' \quad \text{inside}$$

63) The U.S. Postal Service will accept a box for domestic shipment only if the sum the length and girth (distance around, like perimeter) does not exceed 108 inches. What dimensions will give a box with a square end the largest possible volume?



$$V = x^2 \cdot y$$

$$4x + y = 108$$

$$y = 108 - 4x$$



$$V = x^2 (108 - 4x)$$

$$V = 108x^2 - 4x^3$$

$$V' = 216x - 12x^2 = 0$$

$$12x(18 - x) = 0$$



$$x = 0$$



$$x = 18''$$



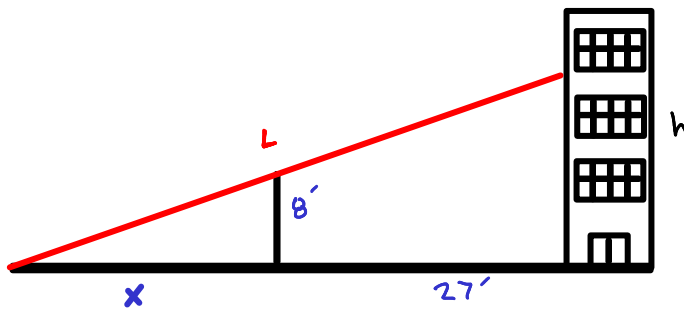
$$y = 108 - 4(18)$$

$$y = 36''$$

$$x = 18''$$

$$y = 36''$$

64) *Shortest Beam.* There is an 8-foot tall wall that stands 27 feet away from a building. Find the length of the shortest straight beam that will reach to the side of the building from the ground outside the wall.



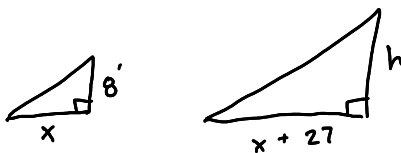
$$h = \frac{8(12 + 27)}{12}$$

$$h = 26'$$

$$L^2 = 26^2 + (12 + 27)^2$$

$$L = \sqrt{26^2 + (39)^2}$$

$$L = 46.872'$$



$$\frac{h}{8} = \frac{x+27}{x} \Rightarrow h = \frac{8(x+27)}{x}$$

$$L^2 = h^2 + (x+27)^2$$

$$L^2 = \frac{64(x+27)^2}{x^2} + (x+27)^2$$

$$L^2 = (x+27)^2 \left[\frac{64}{x^2} + 1 \right]$$

$$2 \cdot L \cdot L' = 2(x+27) \left[\frac{64}{x^2} + 1 \right] + (x+27)^2 \left[\frac{-128}{x^3} \right]$$

↓

$$0 = 2(x+27) \left[\frac{64}{x^2} + 1 + (x+27) \left(\frac{-64}{x^3} \right) \right]$$

↓

$$0 = \cancel{\frac{64}{x^2}} + 1 - \cancel{\frac{64}{x^2}} - \frac{27 \cdot 64}{x^3}$$

$$0 = 1 - \frac{27 \cdot 64}{x^3}$$

$$\sqrt[3]{\frac{27 \cdot 64}{x^3}} = \sqrt[3]{x^3} \Rightarrow$$

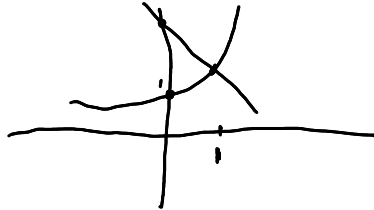
$$\sqrt[3]{27} \sqrt[3]{64} = x$$

$$3 \cdot 4 = x$$

$$x = 12'$$

Section 4.8 Newton's Method

65) Use Newton's Method to estimate the only solution for $e^x = -2x + 3$.



$$f(x) = e^x + 2x - 3$$

$$f'(x) = e^x + 2$$

$$x = \frac{(e^x + 2x - 3)}{(e^x + 2)}$$

$$x_0 = 1$$

$$x_1 = 0.636$$

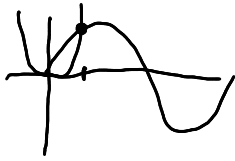
$$x_2 = 0.595$$

$$x_3 = 0.594$$

$$x_4 = 0.594$$

$$\left. \begin{array}{l} x_3 = 0.594 \\ x_4 = 0.594 \end{array} \right] \boxed{x = 0.594}$$

66) Use Newton's Method to estimate the smallest positive solution for $\sin x = x^2$.



$$f(x) = \sin x - x^2$$

$$f'(x) = \cos x - 2x$$

$$x = \frac{\sin x - x^2}{\cos x - 2x}$$

$$x_0 = 1$$

$$x_1 = 0.891$$

$$x_2 = 0.877$$

$$x_3 = 0.877$$

$$\boxed{x = 0.877}$$

Section 4.9 Antiderivatives

67) Find the antiderivative for $g(x) = \frac{1}{x^2} + \frac{1}{x} + \sec^2 x$.

$$y' = \frac{1}{x^2} + \frac{1}{x} + \sec^2 x \rightarrow y = ?$$

$$\int \left(\frac{1}{x^2} + \frac{1}{x} + \sec^2 x \right) \cdot (dx)$$

$$\int \left(\underbrace{x^{-2+1}}_{-1} + \frac{1}{x} + \sec^2 x \right) \cdot (dx)$$

$$= \frac{x^{-1}}{-1} + \ln x + \tan x + C$$

$$\boxed{= -\frac{1}{x} + \ln x + \tan x + C}$$

68) Find $f(x)$ if $f'(x) = \sqrt[3]{x} + 6x^4$ and $f(0) = 2$.

$$y' = \frac{x^{1/3+1}}{4/3} + \frac{6x^{4+1}}{5}$$

$$y = \frac{3}{4} x^{4/3} + \frac{6}{5} x^5 + C$$

$$2 = \frac{3}{4}(0) + \frac{6}{5}(0) + C$$

$$C = 2$$

$$y = \frac{3}{4} x^{4/3} + \frac{6}{5} x^5 + 2$$

69) Find $f(x)$ if $f''(x) = \sin x + \cos x$ and $f'(0) = 1$, $f(0) = 2$.

$$f'(x) = -\cos x + \sin x + C$$

$$1 = -\cos(0) + \sin(0) + C$$

$$1 = -1 + 0 + C$$

$$2 = C \quad \downarrow$$

$$f'(x) = -\cos x + \sin x + 2$$

$$f(x) = -\sin x - \cos x + 2x + K$$

$$2 = -\sin(0) - \cos(0) + 2(0) + K$$

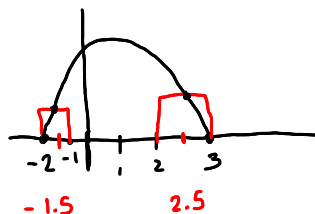
$$2 = 0 - 1 + 0 + K$$

$$3 = K \quad \downarrow$$

$$f(x) = -\sin x - \cos x + 2x + 3$$

Section 5.1 Areas and Distances

70) Consider the region that is above the x -axis and the function $y = -x^2 + x + 6$. Estimate this area as accurately as possible using 5 rectangles.



$$\Delta x = \frac{3 - (-2)}{5} = 1$$

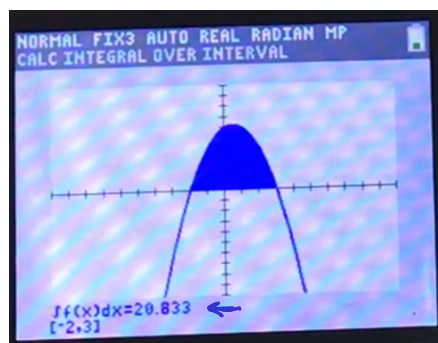
$$\text{Sum}(\text{seq}((-x^2 + x + 6) \cdot 1, x, -1.5, 2.5, 1))$$



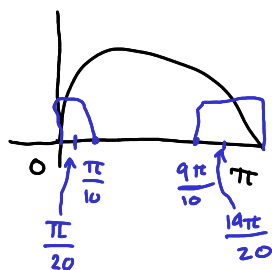
$$= 21.250 \text{ estimate}$$

using 5 rectangles

Actual Answer: 20.833



71) Estimate the total area between $y = \sin x$ on $[0, \pi]$ and the x -axis using 10 rectangles.



$$\Delta x = \frac{\pi - 0}{10} = \frac{\pi}{10}$$

$$\text{Sum} \left(\text{seq} \left(\sin x \cdot \frac{\pi}{10}, x, \frac{\pi}{20}, \frac{19\pi}{20}, \frac{\pi}{10} \right) \right)$$

$= 1.959$ estimated
using 10 rectangles

Actual Answer: 2

